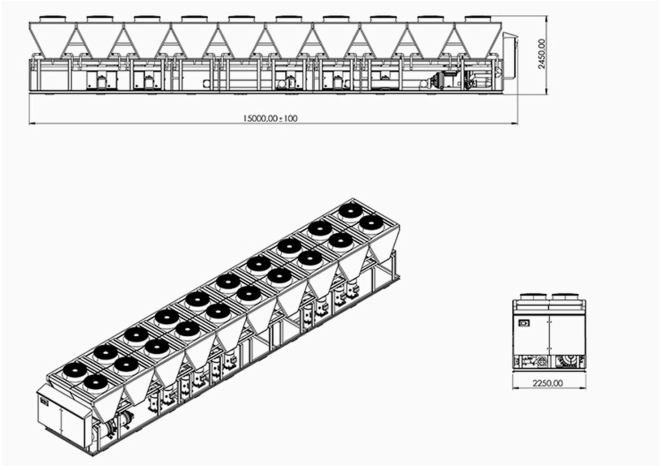


# FPROS1-A1290-OSB-CU

Serie FPROS Cu-Al · R1234ZE · senza serbatoio / without tank



FPROS1-A1290-OSB-CU

RESA FRIGORIFERA · COOLING CAPACITY

840 kW

EER · 12/7 °C · AIR 35 °C

2,89

RESA NOMINALE · 15 °C · AIR 25 °C

1,283 kW

PORTATA ACQUA · WATER FLOW

144,5 m³/h

|                            |                 |                                        |                        |
|----------------------------|-----------------|----------------------------------------|------------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 15.000 × 2.250 × 2.450 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                      |
| Ventilatori / Fans         | 20              | Taglia / Size                          | 1290                   |
| Free cooling (ΔT 8 K)      | 1.000 kW *      | Campo aria esterna / Ambient air range | -5 / +47 °C            |
| Refrigerante / Refrigerant | R1234ZE · GWP 7 |                                        |                        |

## Resa in sintesi / Capacity at a glance kW

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 941   | 892   | 840   | 788   | 738   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.283 | 1.216 | 1.151 | 1.085 | 1.016 |

Refrigeratori con free cooling integrato, compressori a vite e batterie in tubi di rame e alette in alluminio, per grandi potenze che lavorano tutto l'anno.

Chillers with integrated free cooling, screw compressors and copper-tube, aluminium-fin coils, for large capacities running all year round.

# Dati tecnici

## Technical data

FPROS1 - A1290 - OSB - CU

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.283,05 kW              |
| Resa frigorifera / Cooling capacity | 1.103.423 kcal/h         |
| Potenza assorbita / Power input     | 249,38 kW                |
| EER / EER                           | 5,14                     |
| Portata acqua / Water flow          | 220,69 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 839,86 kW                |
| Potenza assorbita / Power input     | 290,61 kW                |
| EER / EER                           | 2,89                     |
| ESEER / ESEER                       | 3,84                     |
| Portata acqua / Water flow          | 144,46 m <sup>3</sup> /h |

### Componenti / Components

|                                               |                         |
|-----------------------------------------------|-------------------------|
| Compressori / Compressors                     | 2 × Vite                |
| Circuiti frigoriferi / Refrigerant circuits   | 2                       |
| Ventilatori / Fans                            | 20                      |
| Free cooling (ΔT 8 K) / Free cooling (8 K ΔT) | 1.000 kW *              |
| Ventilatori free cooler / Free cooler fans    | 20                      |
| Condensatore / Condenser                      | batteria rame-alluminio |
| Evaporatore / Evaporator                      | a fascio tubiero        |
| Refrigerante / Refrigerant                    | R1234ZE • GWP 7 • A2L   |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 15.000 × 2.250 × 2.450 mm |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

\* Potenza di free cooling indicativa, da confermare in fase di offerta. / Indicative free cooling capacity, to be confirmed at quotation stage.

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light
- Batteria rame-alluminio  
Copper-aluminium coil

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 823,1 kW     |
| MEG 30 % | × 0,950 | 797,9 kW     |
| MEG 40 % | × 0,920 | 772,7 kW     |
| MPG 20 % | × 0,975 | 818,9 kW     |
| MPG 30 % | × 0,960 | 806,3 kW     |
| MPG 40 % | × 0,950 | 797,9 kW     |

# Prestazioni Performance

FPROS1-A1290-OSB-CU · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    |
|---------------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 813   | 774   | 734   | 694   | 654   | 612   | 569   |
| 2             | 849   | 807   | 767   | 724   | 683   | 639   | 596   |
| 3             | 884   | 841   | 799   | 754   | 712   | 667   | 622   |
| 4             | 921   | 877   | 834   | 787   | 743   | 697   | 650   |
| 5             | 958   | 913   | 869   | 820   | 773   | 727   | 678   |
| 6             | 997   | 951   | 905   | 856   | 806   | 758   | 708   |
| 7             | 1.037 | 989   | 941   | 892   | 840   | 788   | 738   |
| 8             | 1.080 | 1.030 | 980   | 929   | 876   | 823   | 770   |
| 9             | 1.123 | 1.072 | 1.019 | 966   | 912   | 858   | 802   |
| 10            | 1.168 | 1.115 | 1.061 | 1.004 | 950   | 893   | 835   |
| 11            | 1.213 | 1.158 | 1.102 | 1.043 | 987   | 928   | 869   |
| 12            | 1.260 | 1.203 | 1.146 | 1.086 | 1.027 | 967   | 904   |
| 13            | 1.307 | 1.248 | 1.191 | 1.128 | 1.066 | 1.005 | 939   |
| 14            | 1.358 | 1.297 | 1.237 | 1.172 | 1.109 | 1.045 | 978   |
| 15            | 1.410 | 1.346 | 1.283 | 1.216 | 1.151 | 1.085 | 1.016 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 201 | 219 | 239 | 261 | 286 | 312 | 342 |
| 2             | 202 | 220 | 240 | 262 | 286 | 313 | 343 |
| 3             | 203 | 221 | 241 | 263 | 287 | 314 | 344 |
| 4             | 203 | 222 | 241 | 263 | 288 | 315 | 345 |
| 5             | 204 | 222 | 242 | 264 | 289 | 316 | 346 |
| 6             | 205 | 223 | 243 | 265 | 290 | 317 | 347 |
| 7             | 205 | 224 | 244 | 266 | 291 | 318 | 348 |
| 8             | 206 | 224 | 244 | 266 | 291 | 319 | 349 |
| 9             | 207 | 225 | 245 | 267 | 292 | 320 | 350 |
| 10            | 208 | 226 | 246 | 268 | 293 | 320 | 351 |
| 11            | 209 | 226 | 246 | 269 | 294 | 321 | 352 |
| 12            | 209 | 227 | 247 | 269 | 294 | 322 | 353 |
| 13            | 210 | 228 | 248 | 270 | 295 | 323 | 353 |
| 14            | 211 | 229 | 249 | 271 | 296 | 323 | 354 |
| 15            | 212 | 230 | 249 | 271 | 296 | 324 | 355 |

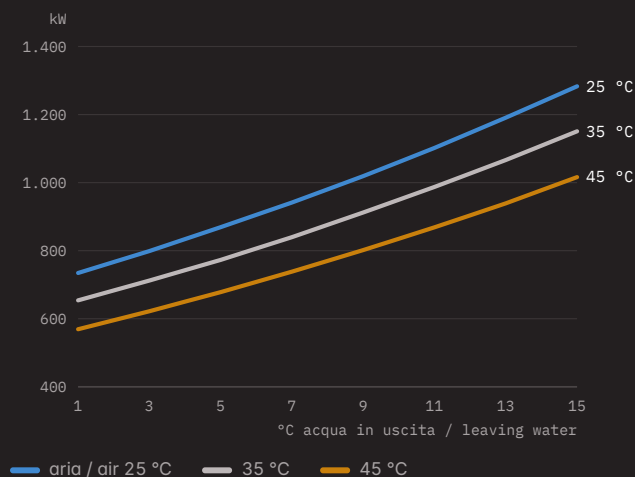
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,04 | 3,53 | 3,07 | 2,66 | 2,29 | 1,96 | 1,66 |
| 2             | 4,20 | 3,67 | 3,19 | 2,77 | 2,38 | 2,04 | 1,74 |
| 3             | 4,36 | 3,81 | 3,32 | 2,87 | 2,48 | 2,12 | 1,81 |
| 4             | 4,53 | 3,96 | 3,45 | 2,99 | 2,58 | 2,21 | 1,88 |
| 5             | 4,69 | 4,11 | 3,59 | 3,10 | 2,68 | 2,30 | 1,96 |
| 6             | 4,87 | 4,27 | 3,73 | 3,23 | 2,78 | 2,39 | 2,04 |
| 7             | 5,05 | 4,42 | 3,87 | 3,36 | 2,89 | 2,48 | 2,12 |
| 8             | 5,24 | 4,59 | 4,01 | 3,49 | 3,01 | 2,58 | 2,21 |
| 9             | 5,43 | 4,76 | 4,16 | 3,61 | 3,12 | 2,68 | 2,29 |
| 10            | 5,62 | 4,94 | 4,32 | 3,75 | 3,24 | 2,79 | 2,38 |
| 11            | 5,82 | 5,11 | 4,47 | 3,88 | 3,36 | 2,89 | 2,47 |
| 12            | 6,02 | 5,30 | 4,64 | 4,03 | 3,49 | 3,00 | 2,56 |
| 13            | 6,22 | 5,48 | 4,80 | 4,18 | 3,61 | 3,11 | 2,66 |
| 14            | 6,44 | 5,67 | 4,98 | 4,33 | 3,75 | 3,23 | 2,76 |
| 15            | 6,65 | 5,86 | 5,14 | 4,48 | 3,88 | 3,35 | 2,86 |

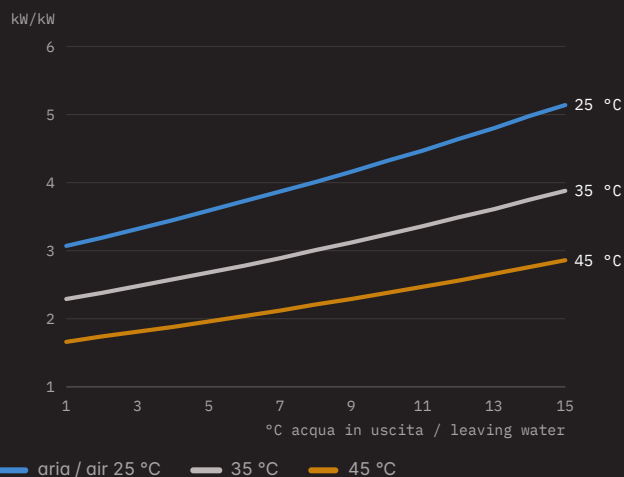
# Curve e dimensioni

## Curves and dimensions

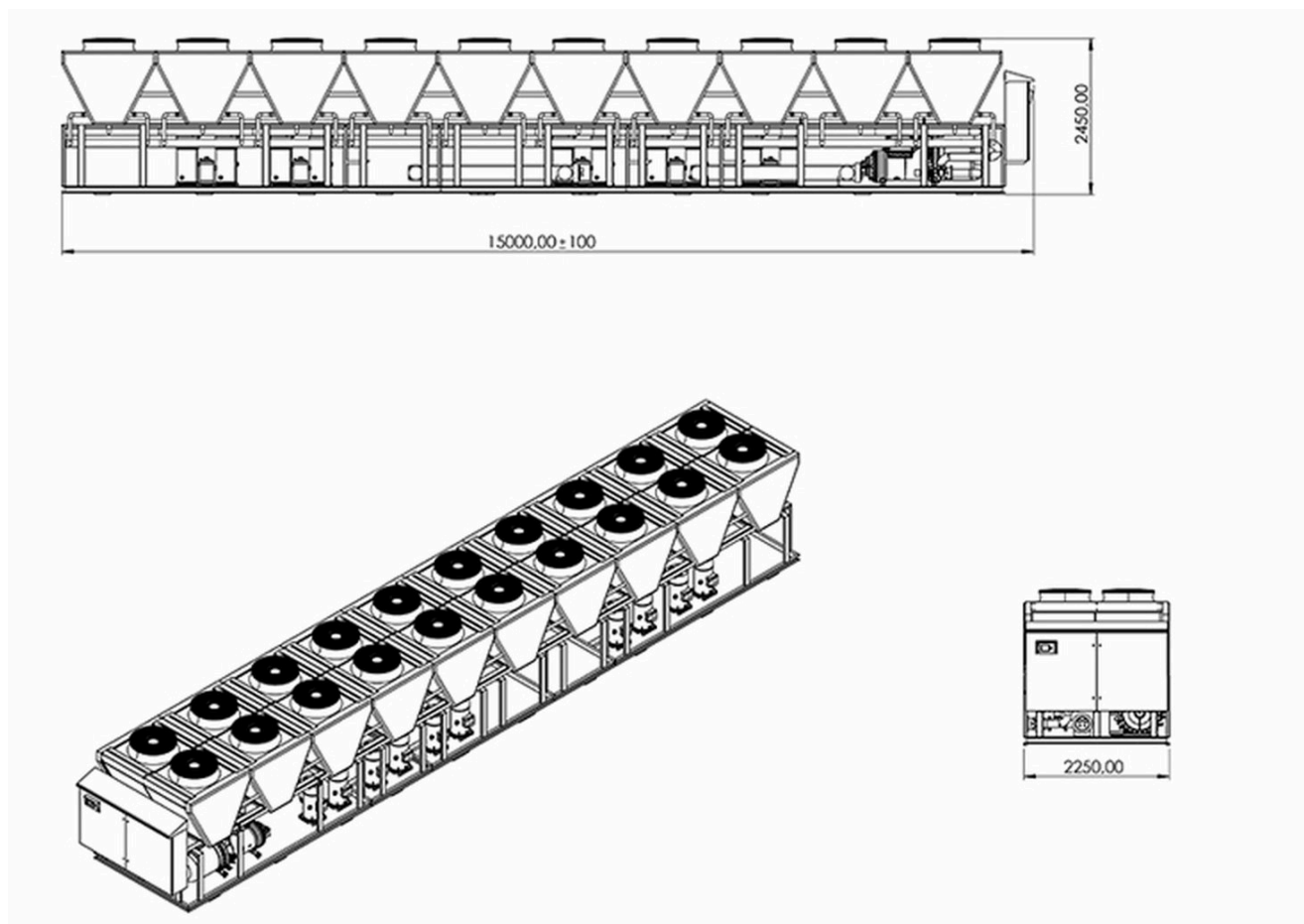
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 15.000 × 2.250 × 2.450 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

FPROS1-A1290-OSB-CU · 59 optional disponibili per questa configurazione · 59 options available for this configuration

## Pompe / Pumps

Pompa 177 m³/h – prevalenza 25 m  
Pump 177 m³/h – head 25 m

Pompa 177 m³/h – prevalenza 40 m  
Pump 177 m³/h – head 40 m

Pompa doppia montata 177 m³/h – prevalenza 25 m  
Twin pump, factory -fitted, 177 m³/h – head 25 m

Pompa doppia montata 177 m³/h – prevalenza 40 m  
Twin pump, factory -fitted, 177 m³/h – head 40 m

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata  
External water tank 3000 l, AISI 304 with factory -fitted twin AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata  
External water tank 5000 l, AISI 304 with factory -fitted twin AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata  
External water tank 3000 l, AISI 304 with factory -fitted twin pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata  
External water tank 5000 l, AISI 304 with factory -fitted twin pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata  
External water tank 2000 l, AISI 304 with factory -fitted single AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata  
External water tank 3000 l, AISI 304 with factory -fitted single AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata  
External water tank 5000 l, AISI 304 with factory -fitted single AISI 304 pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata  
External water tank 2000 l, AISI 304 with factory -fitted single pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata  
External water tank 3000 l, AISI 304 with factory -fitted single pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata  
External water tank 5000 l, AISI 304 with factory -fitted single pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata  
External water tank 3000 l, pressurised, galvanised steel with factory -fitted twin pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata  
External water tank 5000 l, pressurised, galvanised steel with factory -fitted twin pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata  
External water tank 3000 l, pressurised, galvanised steel with factory -fitted single pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata  
External water tank 5000 l, pressurised, galvanised steel with factory -fitted single pump

Vaso di espansione 100 l  
Expansion vessel 100 l

Vaso di espansione 200 l  
Expansion vessel 200 l

Vaso di espansione 300 l  
Expansion vessel 300 l

Vaso di espansione 500 l  
Expansion vessel 500 l

Vaso di espansione 750 l  
Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"  
Solenoid valve 2"

Elettrovalvola 3"  
Solenoid valve 3"

Elettrovalvola 4"  
Solenoid valve 4"

Elettrovalvola 5"  
Solenoid valve 5"

Elettrovalvola 6"  
Solenoid valve 6"

Filtro a dischi in acciaio 2"  
Steel disc filter 2"

Filtro a dischi in acciaio 3"  
Steel disc filter 3"

Filtro a dischi in acciaio 4"  
Steel disc filter 4"

Filtro meccanico a Y 2"  
Y -strainer 2"

Filtro meccanico a Y 3"  
Y -strainer 3"

Filtro meccanico a Y 4"  
Y -strainer 4"

Filtro meccanico a Y 5"  
Y -strainer 5"

Filtro meccanico a Y 6"  
Y -strainer 6"

Valvola a 3 vie 2" miscelatrice  
3 -way mixing valve 2"

Valvola a 3 vie 3" miscelatrice  
3 -way mixing valve 3"

# Optional — 2/2

## Options

FPROS1-A1290-OSB-CU · 59 optional disponibili per questa configurazione · 59 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
  - Free cooling: potenza riferita a una differenza di 8 K tra acqua e aria esterna; valore indicativo, da confermare in fase di offerta.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.
  - Free cooling: capacity referred to an 8 K difference between water and outdoor air; indicative value, to be confirmed at quotation stage.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

Procryon — Smart Cooling Engineering S.r.l.  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
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Continuity by design